

Project Title: Data Access Audit & Insider Threat Detection

Team Name: Horizon

1. Executive Summary

Our solution is an AI-powered Security Operations Dashboard designed to combat insider threats by automating the detection and investigation of 1M+ daily data access events.

2. Platform Architecture

- **Frontend:** Built using React.js with recharts for dynamic data visualization. The interface provides a centralized view of security metrics, compliance status, and user risk levels.
- **Data Ingestion:** The system is engineered to ingest logs from multiple sources (SQL databases, cloud buckets, and BI tools).
- **Processing:** The application uses a React-based runtime environment. It performs real-time anomaly scoring, categorizing events by sensitivity and behavior patterns.
- **Scalability Strategy:** While this prototype uses JSON-based data, the architecture is designed for production-level scale using:
 - **Ingestion:** Apache Kafka for high-velocity log streaming.
 - **Processing:** Apache Spark for distributed compute, ensuring latency remains under 5 minutes.
 - **Storage:** S3 Data Lakes and Parquet files for long-term audit trail storage.

3. Analysis Algorithms

- **Behavioral Anomaly Detection:** The system assigns a risk score (0-100) to every user by analyzing three key vectors:
 1. **Temporal Anomaly:** Accessing data outside of typical "9-to-5" windows (e.g., 2 AM).
 2. **Volume Anomaly:** Bulk data exports exceeding historical user averages.

- 3. **Access Path Anomaly:** Unauthorized cross-departmental data access (e.g., HR personnel accessing Finance databases).
- **AI Narrative Engine:** We integrate the **Llama-3-70b-versatile** LLM via API to translate raw event data into human-readable incident narratives. This allows investigators to instantly understand *why* an event was flagged.

4. UI/UX Design

- **Executive Dashboard:** Provides a high-level view of compliance coverage (GDPR, NIST, SOX) and real-time threat heatmaps.
- **Incident Investigation:** A drill-down view where analysts can see specific flagged events and interact with the AI Security Copilot to ask questions about specific threats.
- **Exportability:** One-click PDF report generation for regulatory auditing and compliance documentation.